HOMEWORK 9.1 · UNIT 9 · NC.8.G.9 · NC.8.EE.1

Volume of Cylinders, Cones, Spheres (Intro)

Targets: G.9-1, G.9-2, G.9-3, G.9-4, G.9-5, EE.1-1, G.9-6, G.9-7, G.9-10, G.9-9

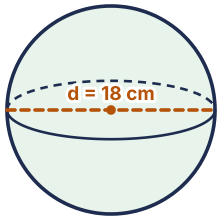
Name _____ Date _____ Block _____

Remember: Cylinder $V = \pi r^2 h$ · Cone $V = \frac{1}{3}\pi r^2 h$ · Sphere $V = \frac{4}{3}\pi r^3$. Radius r = diameter ÷ 2. Volume uses cubic units, like cm^3. **Ex: cylinder** $r = 2$, $h = 5$: $V = \pi(2)^2(5) = 20\pi \approx 62.8\ cm^3$.

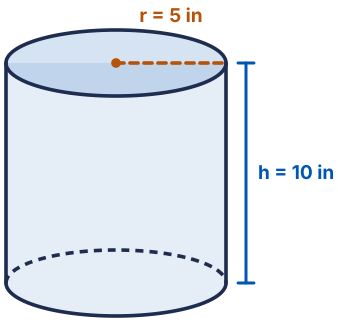
SCORE 3.0 Approaching · the basics

Targets: G.9-1, G.9-2, G.9-3, G.9-4, G.9-5, EE.1-1

1 This ball has a diameter of 18 cm. What radius goes into the formula? G.9-1

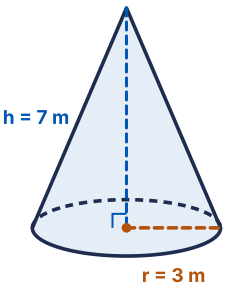


2 Find the volume. Answer in terms of π. G.9-2

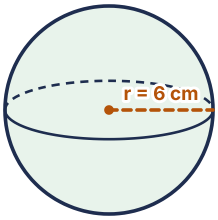


(A) $250\pi\ in^3$ (B) $50\pi\ in^3$
(C) $100\pi\ in^3$ (D) $1{,}000\pi\ in^3$

3 Find the volume of the cone. Answer in terms of π. G.9-3



4 Find the volume of the sphere. Give it in terms of π **and** rounded (3.14, nearest tenth). G.9-4,G.9-5



5 Evaluate 5^3. REVIEW 2.1 EE.1-1

(A) 15 (B) 25 (C) 125 (D) 8

SCORE 4.0 Proficient · on target Targets: G.9-6, G.9-7

6 A cylinder holds 90π ft^3. A cone has the same radius and height. How much does the cone hold? G.9-6

(A) 30π ft^3 (B) 45π ft^3
(C) 270π ft^3 (D) 60π ft^3

7 **Jordan** made a mistake. Circle it. Explain it. Fix it. G.9-7

JORDAN'S WORK
Sphere: $d = 10$ in
$V = \frac{4}{3}\pi(10)^3$
$V = \frac{4{,}000\pi}{3}$ in^3

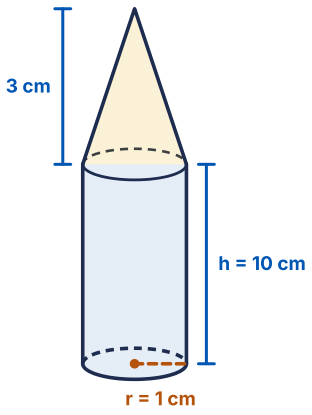
8 A glass is a cylinder with radius 4 cm and height 15 cm. How much water can it hold? Use 3.14 for π. Round to the nearest tenth. G.9-7

SCORE 5.0 Mastery · go deeper Targets: G.9-10, G.9-9

9 Cylinder A has r = 3 and h = 4. Cylinder B has r = 3 and h = 8. Compare their volumes. What would happen if you doubled the radius instead? G.9-10

B holds _____ times as much. Doubling the radius _____ .

10 A pencil is a cylinder with a cone tip. Find its volume. Answer in terms of π. Then round. Show each step. G.9-9



Cylinder = _____ . Cone = _____ . Total = _____ .